PRACTICAL 5 EVIDENCE PRESERVATION & ANALYSIS

1. Objective

The objective of this practical is to demonstrate digital forensic evidence collection, preservation, and integrity verification. This includes capturing volatile system data, creating evidence files, generating cryptographic hashes, and maintaining a chain-of-custody record.

2. Tools Used

- Linux Terminal (Kali Linux)
- netstat utility for volatile data collection
- SHA256 hashing utility (sha256sum)
- Excel / Google Sheets for documentation
- Velociraptor (conceptual reference for forensic collection workflow)

3. Volatile Data Collection (Network Connections)

Command Executed

netstat -ano

Observations

The system displayed active network connections and listening services:

- TCP services listening on ports: 443, 1514, 1515, 55000
- Local loopback activity (127.0.0.1)
- DHCP communication between 192.168.0.110 and 192.168.0.1
- Active UNIX domain socket communications for system processes

Interpretation

The presence of multiple listening services indicates normal system or security monitoring activity. No suspicious external malicious connections were observed during the capture. This represents a stable system state at the time of volatile data acquisition.

4. Evidence File Creation

A sample forensic evidence file was created to simulate data preservation for hashing and integrity verification.

- **Command Used**

echo "SOC Practical 5 Evidence File - Forensic Integrity Check" > sample.txt

5. Evidence Integrity Verification (SHA256 Hashing)

Command Executed

sha256sum sample.txt

Output

0259f93f08965287de83559e73e715e06dc687ab23673fff05e2d7a0a68b312a sample.txt

Interpretation

The SHA256 hash represents a unique digital fingerprint of the evidence file. Any modification to the file would result in a completely different hash value, ensuring integrity verification during forensic investigations.

6. Chain of Custody Record

Evidence ID	Evidence Type	Description	Collected By	Date	Hash Value
E-001	Volatile Data	Netstat network connections	SOC Analyst	2026-01-13	N/A
E-002	File Evidence	sample.txt forensic file	SOC Analyst	2026-01-13	0259f93f08965287de83559e73e715e06dc687ab23673fff05e2d7a0a68b312a

7. SOC Relevance

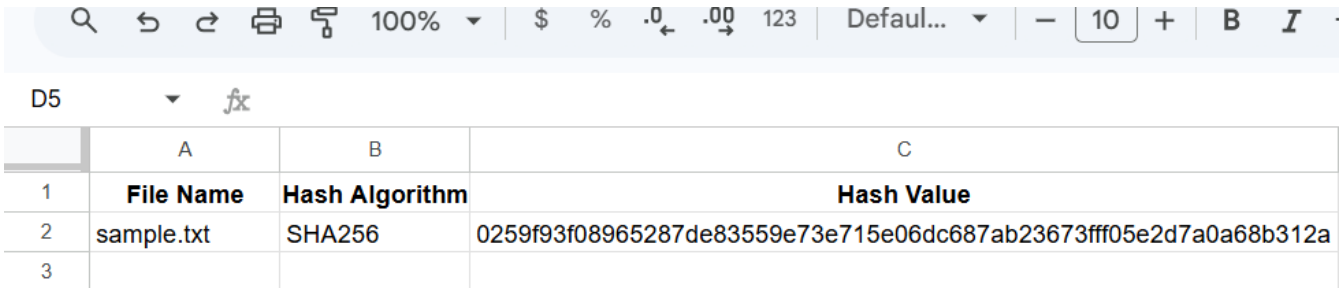
This practical demonstrates key digital forensic principles:

- Collection of volatile system data
- Preservation of digital evidence
- Cryptographic integrity verification
- Chain-of-custody documentation
- Forensic readiness in SOC environments

8. Conclusion

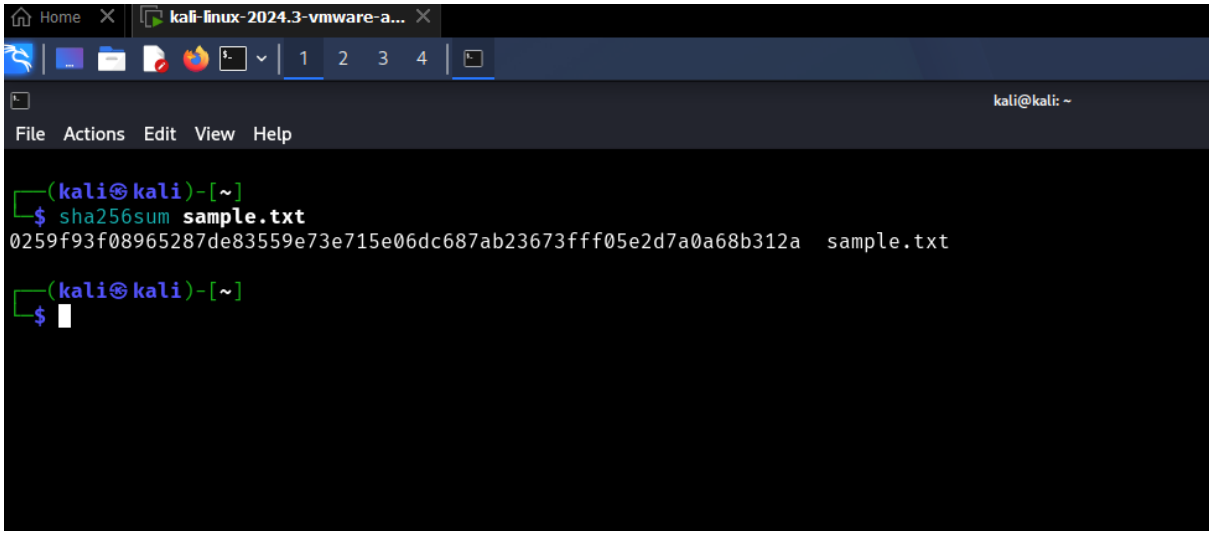
Evidence preservation was successfully demonstrated through volatile data capture and file-based integrity verification. The use of SHA256 hashing ensures tamper detection, while chain-of-custody documentation maintains forensic accountability. This workflow reflects standard SOC and digital forensic investigation procedures.

9.Screenshots



D5

	A	B	C
1	File Name	Hash Algorithm	Hash Value
2	sample.txt	SHA256	0259f93f08965287de83559e73e715e06dc687ab23673fff05e2d7a0a68b312a
3			



D9

	A	B	C	D
1	**Item**	**Description**	**Collected By**	**Date**
2	Memory Dump	Server-Y Dump	SOC Analyst	2026-01-13

F6

	A	B	C	D
1	**Protocol**	**Local Address**	**Foreign Address**	**State**
2	TCP	192.168.1.100:4	8.8.8.8:443	ESTABLISHED
3	TCP	192.168.1.100:8	45.33.32.156:80	CLOSE_WAIT

H6

	A	B	C	D	E
1	**Evidence Type**	**Description**	**Collected By**	**Date**	**Hash Value**
2	Netstat Log	Active connectio	SOC Analyst	2026-01-13	N/A
3	Memory Dump	Server-Y dump	SOC Analyst	2026-01-13	SHA256 value